Coverage Validation: `edgar-crawler` vs Hoberg-Phillips TNIC

Marco Zhang

February 2026

1 Current Stage & Goal

In the previous report, we explored `edgar-crawler`'s mechanics: how it downloads filings from EDGAR, parses raw HTML/txt, and extracts item-level text. We documented configuration caveats (exact filing-type matching misses 47% of variants, item lists need manual upkeep across SEC eras) and confirmed extraction reliability across 1993–2024.

The next question: **how does edgar-crawler's coverage compare to an established benchmark?** We choose Hoberg-Phillips TNIC (2005) because (1) it is widely used in finance/accounting research, (2) it is built from 10-K Item 1 extractions—the same task `edgar-crawler` performs, and (3) it publishes both data (pairwise similarity scores by gvkey-year) and replication code (Python notebooks). This lets us validate at two levels: file-level (do both tools see the same firms?) and item-level (do both tools extract the same text?).

2 File-Level Coverage

TNIC uses Compustat gvkeys—its universe is public companies traded on the three major US exchanges. EDGAR uses CIK as identifier, covering all SEC registrants (small business filers, foreign issuers, shell companies), so it is much broader.

We match EDGAR filings to TNIC firms using the WRDS `wciklink_gvkey` crosswalk (historical CIK-gvkey mapping from the SEC Analytics Suite). We filter EDGAR filings by `year(datadate) == 2005`, which exactly matches TNIC's year definition—confirmed from H-P's own README[1]: *"the year field in this database is based on Compustat calendar years obtained as the first four digits of the YYYYMMDD datadate variable."*

[1] `https://hobergphillips.tuck.dartmouth.edu/idata/Readme_tnic3HHIData.txt`

Method	FY2005 CIKs	TNIC Covered	Rate
SEC API + pagination	12,038	5,084 / 5,122	**99.3%**

Table 1: File-level coverage: EDGAR vs TNIC (2005).

This is a strict, exact-fiscal-period match—not a loose filing-window heuristic—making 99.3% a reliable statistic. Good enough to proceed from file-level to item-level comparison.

2.1 Filing Type Breakdown

Among the 5,137 TNIC-matched CIKs with FY2005 filings, **∼94% file 10-K and ∼6% file 10KSB**:

Filing Type	TNIC-Matched Firms	%
10-K / 10-K/A	4,824	93.9%
10KSB / 10KSB/A	313	6.1%

Table 2: Filing type distribution among TNIC-matched FY2005 firms (Compustat universe).

This tells us that for pre-2009 data (before the SEC eliminated the 10KSB form), we need parsing methodology that handles 10KSB filings—not just 10-K.

If we want to use the **full EDGAR CIK universe** (not just TNIC-matched), the 10KSB share is even larger:

Filing Type	Count	Share
10-K	9,003	59.9%
10KSB	3,223	21.5%
10-K/A	1,517	10.1%
10KSB/A	1,262	8.4%
10-KT / 10-KT/A	19	0.1%
10-K variants	**10,520**	**73.7%**
10KSB variants	**4,485**	**26.2%**

Table 3: FY2005 filing type breakdown across all EDGAR CIKs.

So for full-universe coverage, we need parsing methodology for more than just 10-K and 10KSB—amendment variants (10-K/A, 10KSB/A) and transition reports (10-KT) also need handling.

2.2 The Last 0.7%

We diagnosed all 38 NOT_COVERED gvkeys. The largest category (17/38 = 45%) is **SEC API data quality issues**—the filings exist in EDGAR, but the SEC API returns an incorrect `reportDate` (equal to the filing date, or off by one day), so our year-matching filter misses them.

Category	Count	Key Insight	Example
SEC rd ≈ filingDate	14	Dec FY-end; SEC returns rd = fd	Ball Corp
SEC rd off-by-one	3	rd = 2006-01-01 not 2005-12-31	Interface Inc
Genuine non-Dec FY	6	52/53-wk years or Jan/Mar FY	Yum Brands
Missing FY2005 filing	5	No 10-K for FY2005 in EDGAR	Nature's Sunshine
Late filer	4	FY05 10-K filed years late	Am. Italian Pasta
Spinoff / new entity	4	CIK didn't exist until 2006+	Celera Corp
Foreign private issuer	1	Files 6-K only, not 10-K	AXA S.A.
No crosswalk	1	No wciklink entry	gvkey 62642
Total	**38**		

Table 4: Diagnosis of all 38 NOT_COVERED gvkeys (0.7% gap).

3 Item-Level Extraction

H-P only extract Item 1 (Business Description) and only provide replication code for that item, so we focus our comparison on Item 1. We use `edgar-crawler`'s 62 test fixture 10-K filings (spanning 1993–2023, mixed `.txt`/`.htm` formats) and run both parsers on each filing:

- **edgar-crawler**: `ExtractItems` class with `items_to_extract=["1"]`, `lxml` parser, newline-anchored regex.
- **H-P**: reimplemented from their Notebook 2—4-regex cascade, BeautifulSoup `html.parser`, longest-match disambiguation, 2KB minimum filter.

We compare the extracted Item 1 text using word-level Jaccard similarity.

3.1 Results

Extraction success rate:

- edgar-crawler: **62/62 (100%)**

- H-P: **35/62 (56.5%)**

Jaccard similarity (35 filings where both extracted): Mean = 0.910, Median = 0.946, Min = 0.241, Max = 0.999.

Category	Count	Share	Description
MATCH	16	16/35	Jaccard $\geq$ 0.95, essentially identical
BOUNDARY_DIFF	17	17/35	Jaccard 0.70–0.95, same core, different boundaries
MAJOR_DIFF	2	2/35	Jaccard < 0.70, substantially different
EC_ONLY	27	—	edgar-crawler extracted, H-P did not
HP_ONLY	0	—	H-P extracted, edgar-crawler did not
BOTH_EMPTY	0	—	Neither extracted anything

Table 5: Item 1 extraction comparison across 62 test fixture filings.

3.2 Key Findings

1. H-P fails on all pre-2004 filings (27 EC_ONLY). H-P's regex looks for `item 1 ... item 1a/1b` boundaries, but many older filings go directly from "ITEM 1. BUSINESS" to "ITEM 2. PROPERTIES"—no Item 1A "Risk Factors" (not mandatory until the 2005 SEC rule change). H-P cannot find an ending boundary, so it returns nothing. edgar-crawler handles this by trying all subsequent items as end boundaries (1A $\rightarrow$ 1B $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow \ldots \rightarrow$ SIGNATURE).

2. H-P includes Table of Contents pollution (BOUNDARY_DIFF). In 15 of 17 BOUNDARY_DIFF cases, H-P's extraction starts earlier—capturing "Item 1" from the Table of Contents and extending to "Item 1A" in the body. edgar-crawler avoids this via newline-anchored regex (`\n` at start of item header), which filters out inline ToC references. For modern iXBRL filings (2019+), H-P also captures raw HTML/CSS because `BeautifulSoup('html.parser')` doesn't fully parse inline XBRL tags.

3. Two MAJOR_DIFF cases.

- Horizon Financial 2009 (J=0.24): H-P extracted 15KB (ToC fragment only), edgar-crawler extracted 139KB (full Item 1).
- FedEx 2023 (J=0.34): H-P extracted 530KB (raw HTML including CSS/iXBRL), edgar-crawler extracted 112KB (clean text).

4. When both succeed, they agree closely. For the 33 filings with Jaccard $\geq$ 0.70, the mean Jaccard is 0.946. Remaining differences are boundary choices (H-P includes the trailing "Item 1A" header) and table removal (edgar-crawler strips numerical tables).

5. Year distribution confirms the pattern is era-driven.

Era	MATCH	BOUNDARY_DIFF	MAJOR_DIFF	EC_ONLY	Total
1993–1999	0	0	0	14	14
2000–2004	1	0	0	9	10
2005–2010	3	5	1	3	12
2011–2023	12	12	1	1	26

Table 6: Extraction comparison by era. Pre-2005: H-P extracts almost nothing (23/24 EC_ONLY). Post-2010: both work on 24/26 filings (92%).

Pre-2005: H-P extracts almost nothing—no Item 1A boundary available. 2005–2010: transition period, H-P starts working. 2011–2023: both work on 24/26 filings (92%); BOUNDARY_DIFFs are consistently ToC pollution. The 2005 cutoff aligns with SEC's mandatory Item 1A (Risk Factors) requirement.